Blind Adjudication Protocols and Coverage-Aware Risk Bounds

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Abstract

This article specifies a protocol for external comparative adjudication of the operational certificates developed in BaseCore and Papers I–IX. The analytic risk, coverage, and Gaussian-comparison theorems are mathematical protocol results; no executed campaign results are reported. The protocol separates internal certificates from admissible human-comparator evidence, defines preregistration freeze and amendment rules, and specifies independent reviewer selection, blinding, matched interventions, negative controls, and coverage-aware decision rules. Execution-environment integrity checks, isolated reproducibility, dependency specification, deterministic record regeneration, and path-sanitization controls are prerequisites to admissible collection, rather than evidence of the comparative outcome. Explicit gates distinguish rehearsal, session zero, and the first admissible collection event. The article reports no human comparator results or external adjudication verdicts and makes no claim of human equivalence, superiority, inferiority, cross-substrate identity preservation, metaphysical subjecthood, or final comparative conclusions.

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Scope and Operational Postulate

This article studies the stated mathematical objects and their operational realization conditions. Whether a sufficiently realized artificial or biological system has phenomenal experience remains open as an independent bridge hypothesis. Empirical interpretation requires identified measurements and admissible comparisons.

1 Formal Dependency Stack

The external adjudication stack is layered across progressive protocol tiers. These tier identifiers denote formal dependencies in the protocol design; listing them does not certify that any operational gate has been executed or passed:

1. **BaseCore**: active mathematical base layer.
2. **Papers I–IX**: downstream theoretical and methodological papers.
3. **Subjectivity and Bridge Interface**: upstream operational subjectivity and phenomenal-bridge boundary criteria (Papers VIII–IX).
4. **adjudication design (Adjudication Architecture)**: external adjudication architecture and formal protocol specification.
5. **comparator contracts (Comparator Contracts)**: human comparator instantiation contracts and preregistration templates.
6. **independence conditions (Independence and Rigor)**: reviewer independence, blinding criteria, and protocol execution readiness.
7. **preregistration and packaging (Freeze and Packaging)**: cryptographic preregistration freeze architecture, reviewer-selection schemas, adjudicator-assignment schemas, and pilot execution packaging.
8. **provenance reconciliation (Dry-Run and Provenance Closure)**: bounded execution-pilot dry-run logic, provenance reconciliation, and empty typed record schemas.
9. **rehearsal protocol (Rehearsal Protocol)**: rehearsal verification protocol, dry-run event grammar, and rehearsal readiness gating.

10. **collection preparation (Controlled Execution Opening and Environment Integrity)**: session-zero initialization, pre-collection admissibility gating, unpopulated operational shells, isolated reproducibility, path-sanitization checks, and deterministic record regeneration.

11. **collection opening conditions (First Admissible Collection Gate)**: opening criteria, collection-start authorization grammar, and pre-collection opening gate.

No later protocol tier may borrow empirical closure from an earlier tier, and no earlier formal tier may be retroactively interpreted as supplying later execution-era evidence.

2 Theoretical-vs-Protocol Authority Boundary

Paper 10 sits on a strict separation between mathematical formalization and empirical execution:

- BaseCore and Papers I–IX supply mathematical definitions, structural obstructions, finite constructive certificates, and conditional falsification criteria.
- The external adjudication protocol defined in Paper 10 specifies how future empirical campaigns would be preregistered, blinded, audited, and judged.
- Inclusion of this protocol monograph in the collected theoretical framework supplies zero execution-era human data and establishes no empirical support.
- the full protocol specification specify procedural, freeze, rehearsal, and controlled-opening conditions that remain non-evidential until executed independently with external participants.

Therefore publication within the theoretical corpus cannot be substituted for empirical adjudication, and protocol completeness cannot be substituted for experimental closure.

3 Lineage and Provenance Semantics

The preregistration and packaging and provenance reconciliation enforce a strict authority hierarchy; every admissible record must distinguish among:

1. public theoretical framework authority,
2. local protocol specification authority,
3. frozen preregistration authority,
4. reviewer-package authority,
5. adjudicator-package authority, and
6. execution-era evidence authority.

Archived records whose lineage reflects prior protocol snapshots are preserved intact for provenance; live specification bindings are updated explicitly without altering historical records retroactively.

4 Provenance Closure and Snapshot Reconciliation

The provenance reconciliation specifies a formal provenance-closure classification. A recorded specification state may be classified as:

- `exact_head_aligned`;
- `artifact_forward_compatible`;
- `stale_and_regenerated`;
- `historical_snapshots_intentionally_preserved`.

This distinction matters because preregistration and schema manifests are sealed prior to the archival snapshot that records them, allowing deterministic regeneration against the declared baseline without conflating forward-compatible manifests with empirical staleness.

5 External Adjudication Problem Statement

The external adjudication problem goes beyond internal model coherence and specifies:

- what independent reviewers may legitimately inspect;
- what human comparator evidence classes may later become admissible;
- what negative controls must exist before any comparative interpretation;
- how preregistration becomes frozen;
- how reviewer and adjudicator layers become instantiable without identity leakage;
- how dry-run validation differs from evidential execution.

6 Formal Mathematical Grammar of External Adjudication

The adjudication surface is a measurable protocol object, not an impressionistic comparison between substrates. Let $(\mathcal{T}, d_{\mathcal{T}})$ be the metric space of finite execution traces after canonical serialization. A trace contains the ordered event stream, de-identified channel payloads, runtime manifests, negative-control outcomes, and reviewer-facing masks. Admissible evidence lives in the Borel measurable space

$$(\mathcal{T}, \mathcal{B}(\mathcal{T})), \tag{1}$$

where $\mathcal{B}(\mathcal{T})$ is the Borel σ-algebra generated by open $d_{\mathcal{T}}$-balls.

Definition 6.1 (External adjudication object)**.** *An external adjudication object is the tuple*

$$\mathfrak{E} = (P, D_h, D_m, C_n, C_a, \Sigma, \mathcal{R}, \mathcal{A}), \tag{2}$$

where P is a frozen preregistered protocol, D_h and D_m are human and machine trace datasets in $\mathcal{T}$, C_n and C_a are negative and active control families, $\Sigma = \{\sigma_k\}_{k=1}^{K}$ is a finite family of predeclared measurable statistics $\sigma_k : \mathcal{T}^n \to \mathbb{R}$, $\mathcal{R}$ is the reviewer layer, and $\mathcal{A} : \mathcal{Y} \to \{0, 1, \texttt{undetermined}\}$ is the final adjudicator operator on masked evidence summaries $\mathcal{Y}$.

Definition 6.2 (Admissible comparative evidence)**.** *A dataset pair (D_h, D_m) is admissible under $\mathfrak{E}$ only if the following predicates are all true:*

1. *D_h, D_m are measurable finite samples from $(\mathcal{T}, \mathcal{B}(\mathcal{T}))$.*

2. *The preregistered protocol hash, control manifest hash, and analysis shell hash match their frozen records.*

3. *Human-side records are not generated, rewritten, or completed by synthetic systems:* $\epsilon_{\text{syn}} = 0$ *for the human channel.*

4. *Blinding leakage is bounded by the preregistered budget:* $\mathbb{P}(\text{leak}) \leq \epsilon_{\text{blind}}$.

5. *Every declared negative control has an executed and timestamped outcome under the same admissibility grammar.*

Theorem 6.3 (Non-transitivity of internal support)**.** *Let* $\mathcal{G}_{\text{internal}} \in [0,1]$ *denote any internal runtime support score computed from operational invariants, and let* $\mathcal{G}_{\text{external}} \in \{0, 1, \texttt{undetermined}\}$ *denote the external adjudication verdict. In the absence of an admissible measurable dataset pair* (D_h, D_m)*, one has*

$$\mathcal{G}_{\text{internal}} \not\Longrightarrow \mathcal{G}_{\text{external}}. \tag{3}$$

Proof. The internal score is a function of runtime traces and invariant certificates inside the machine channel. The external verdict is a function of masked comparative summaries built from both D_h and D_m, plus reviewer and adjudicator records. If D_h is absent, non-admissible, unblinded, or unfrozen, then the domain condition of $\mathcal{A}$ is not satisfied. Hence the external verdict is either undefined or `undetermined` regardless of the value of $\mathcal{G}_{\text{internal}}$. Therefore internal operational support is not a logical premise from which external comparative adjudication can be inferred. □

Proposition 6.4 (Operational closure of a comparative program)**.** *The comparative program reaches operational closure only when there exists a predeclared alignment map* $T : \mathcal{T} \to \mathcal{Z}$ *into a common Polish feature space* $(\mathcal{Z}, d_{\mathcal{Z}})$ *and the total divergence*

$$\Delta_{\text{comp}} = \lambda_1 W_1(T_{\#}\widehat{\mu}_h, T_{\#}\widehat{\mu}_m) + \lambda_2 D_{\text{KL}}(\widehat{p}_h \| \widehat{p}_m) + \lambda_3 \max_k |\sigma_k(D_h) - \sigma_k(D_m)| \tag{4}$$

is defined and satisfies the frozen budget $\Delta_{\text{comp}} \leq \epsilon_{\text{budget}}$*. This closure is a precondition for comparative interpretation, not a claim of consciousness, identity, or substrate equivalence.*

7 Frozen Preregistration Architecture

The preregistration and packaging introduces a cryptographic freeze architecture extending preregistration templates. The governing schemas (summarized in Appendix A) comprise:

- the frozen preregistration record schema;
- the preregistration freeze policy;
- the preregistration amendment policy;
- the preregistration lock manifest.

The architecture formalizes:

- record classes;
- immutable fields;
- amendment classes;

- freeze provenance chains;
- local-only freeze state;
- explicit external filing pending state.

Definition 7.1 (Cryptographic freeze map)**.** *Let $\mathcal{P}_{\text{config}}$ be the set of canonical preregistration configurations after deterministic serialization. A freeze map is a collision-resistant hash function*

$$H : \mathcal{P}_{\text{config}} \to \{0,1\}^{256}. \tag{5}$$

A preregistration record p is locally frozen at digest h when $H(p) = h$, h is written into the lock manifest, and every amendment policy references h rather than an editable natural-language description.

Theorem 7.2 (Hash-anchored non-alterability)**.** *Assume the hash function H is collision resistant over the canonical serialization domain. If a post-execution preregistration configuration p' has the same published digest as the frozen pre-execution configuration p, then either $p' = p$ byte-for-byte under canonical serialization or a collision for H has been found.*

Proof. By freeze definition, the published digest is $h = H(p)$. If the later record p' also satisfies $H(p') = h$ while $p' \neq p$, then p and p' form a distinct pair with the same hash output. That is precisely a collision in H. Under the collision-resistance assumption, such a pair is computationally infeasible. Thus digest equality supplies evidence of non-alteration of the serialized protocol, modulo the stated cryptographic assumption. □

8 Local-vs-External Freeze Distinction

The local freeze layer must not be conflated with external filing.

Surface	Current state	Interpretation
Template-only preregistration	`preregistration_ready_for_freeze`	structure exists but no lock exists
Local freeze readiness	`local_freeze_ready`	lock architecture exists but no external filing is claimed
Locally frozen record	blocked	would still require explicit lock instantiation
Externally filed record	absent	cannot be claimed from local architecture alone

9 Instantiated-Empty Record Layer

The provenance reconciliation moves beyond package-level specifications by defining typed-but-empty records that remain explicitly non-evidential (Appendices A–B):

- local frozen-record template (Appendix A)
- reviewer candidate-pool template (Appendix A)
- adjudicator candidate-pool template (Appendix A)
- blinding assignment template (Appendix A)
- execution session template (Appendix B)

- decision record template (Appendix B)
- reproducibility package manifest (Appendix B)
- collection-log schema (Appendix B)
- evidence insertion manifest (Appendix B)

These schemas are intentionally empty, typed, and freeze-capable. They represent neither populated evidence, externally filed records, nor reviewer or adjudicator decisions.

10 Reviewer Candidate and Adjudicator Candidate Architecture

The preregistration and packaging introduces typed candidate schemas without instantiating participant identities. The primary components are:

- reviewer candidate schema (Appendix A)
- adjudicator candidate schema (Appendix A)
- reviewer selection workflow (Appendix A)
- adjudicator assignment workflow (Appendix A)

The candidate layer defines qualification summaries, independence assertions, conflict disclosure references, blinding eligibility classes, and explicit `not_started` states.

11 Reviewer Rejection Logic

Section status. protocol specified.

The reviewer package is incomplete unless rejection logic exists. The rejection registry now includes:

- qualification mismatch;
- disclosed conflict;
- undisclosed conflict;
- prior unblinded access;
- insufficient methodological competence;
- insufficient lineage literacy;
- insufficient control/admissibility literacy.

This logic is not yet a reviewer decision. It is the typed boundary that stops future selection from being arbitrary or retrofitted.

12 Blinding Assignment Logic

Section status. protocol specified.

13 Blinding Assignment Logic

Blinding is formalized via a typed assignment manifest rather than informal instruction. The preregistration and packaging defines blinding classes including:

- `fully_blinded`
- `artifact_blinded`
- `control_family_blinded`
- `unblinding_not_authorized`

The key constraint is that no later dry-run or execution stage may silently cross into unblinded interpretation.

14 Human Comparator Instantiation Logic

The comparator contracts establishes the gap between abstract channels and typed collection routes, while preregistration and packaging extends that structure toward freeze-compatible execution readiness:

- every channel binds a human-side artifact family;
- every channel binds a machine-side homologous artifact family;
- every channel binds minimum metadata, controls, and rejection conditions;
- every channel remains blocked from result claims while data is absent.

Definition 14.1 (Homologous channel statistic)**.** *For each channel c, let $T_c : \mathcal{T} \to \mathcal{Z}_c$ be a frozen feature map into a metric standard Borel space $(\mathcal{Z}_c, d_c)$. Given human and machine samples D_h^c, D_m^c, define empirical pushforward measures*

$$\widehat{\mu}_h^c = (T_c)_{\#} D_h^c, \qquad \widehat{\mu}_m^c = (T_c)_{\#} D_m^c.$$

The channel is statistically comparable only if both empirical measures are defined, the feature map was frozen before execution, and the channel-specific missingness policy accepts both samples.

Definition 14.2 (Divergence shell)**.** *The predeclared divergence shell for channel c is*

$$\Delta_c = \alpha_c W_1(\widehat{\mu}_h^c, \widehat{\mu}_m^c) + \beta_c D_{\mathrm{KL}}(\widehat{p}_h^c \| \widehat{p}_m^c) + \gamma_c \| \widehat{s}_h^c - \widehat{s}_m^c \|_1, \tag{6}$$

where W_1 is the Kantorovich–Rubinstein Wasserstein-1 distance, D_{KL} is the Kullback–Leibler divergence on a frozen finite partition or smoothed density model, and $\widehat{s}_h^c, \widehat{s}_m^c$ are predeclared summary vectors. The weights $\alpha_c, \beta_c, \gamma_c \geq 0$ must be frozen before execution.

Proposition 14.3 (Comparator homologation is conditional)**.** *A human–machine channel pair is homologated by Paper 10 only if every term in Δ_c is defined and the frozen channel budget $\Delta_c \leq \epsilon_c$ is satisfied. Homologation is therefore a statistical transport condition over predeclared observables, not a statement that the channels share a biological, phenomenal, or metaphysical substrate.*

Proof. The definition of Δ_c depends on a frozen feature map, empirical pushforward measures, a KL domain, and summary vectors. If any component is missing, the divergence shell is undefined and no homologation predicate can be evaluated. If all components are defined, the only admissible positive judgment is the preregistered inequality $\Delta_c \leq \epsilon_c$. The conclusion is therefore exactly statistical and conditional. □

15 Channel-by-Channel Execution Contracts

15.1 Self-Report Channel

The self-report channel requires prompt-version freezing, missingness policy, de-identification, and explicit rejection of synthetic or relabeled self-report.

15.2 Intervention-Response Channel

The intervention-response channel requires matched intervention and matched negative-control families, dose policy, stop rules, and branch accounting.

15.3 Temporal-Structure Channel

The temporal-structure channel requires a declared segmentation schema, ordering policy, and aggregation policy.

15.4 Valence-Topology Channel

The valence-topology channel requires a shared topology schema and excludes scalar-only substitutes.

15.5 Embodiment-Boundary Channel

The embodiment-boundary channel requires an environment manifest, confound ledger, and error-budget policy.

15.6 Ambiguity-Load Channel

The ambiguity-load channel requires a task-family contract, annotation schema, and annotation reliability policy.

15.7 Transport-Preservation Channel

The transport-preservation channel remains structurally distinct. It requires a real source-target pairing manifest and transport-axis budget binding. The preregistration and packaging defines the pairing shell; it does not populate the pairing.

16 Matched Intervention Dataset Architecture

Matched-intervention architecture is carried by channel contracts and the pilot bundle shell (Appendix B). The bundle remains unpopulated, so the schema exists independently of empirical data.

17 Negative Control Dataset Architecture

Negative-control architecture is similarly carried by channel contracts and the negative-control bundle schema (Appendix C). Controls form an integral part of any future claim surface rather than a post hoc check.

18 Preregistration Architecture

Paper 10 distinguishes three preregistration layers:

1. template surface;
2. local freeze surface;
3. external filing surface.

The present specification defines the schema grammar for the first two; independent external filing remains unexecuted.

19 Adjudication Blind Protocol

Let Y denote the masked evidence bundle and $\ell \in \{h, m\}$ its source label. The reference procedure uses forced choice with a balanced source distribution. A protocol permitting abstention fixes its cost and minimum coverage before evaluating held-out bundles. The randomization of a decision rule is independent of the source conditional on its permitted input.

Definition 19.1 (Blind adjudicator loss)**.** *For $a(Y) \in \{h, m, \texttt{tie}\}$ and a declared $\gamma \in [1/2, 1]$, put*

$$L_\gamma(a(Y), \ell) = \mathbf{1}\{a(Y) \neq \ell,\ a(Y) \neq \texttt{tie}\} + \gamma \mathbf{1}\{a(Y) = \texttt{tie}\}.$$

Report $R_\gamma = \mathbb{E}L_\gamma$, coverage $C = \Pr(a(Y) \neq \texttt{tie})$, and error conditional on coverage when $C > 0$. For acceptance require a preregistered $C \geq C_{\min} > 0$. Forced choice has $C = 1$; always abstaining has $C = 0$ and $R_\gamma = \gamma$.

Theorem 19.2 (Null error lower bound)**.** *Let E denote leakage and $p = \Pr(E)$. If in the clean stratum ℓ is balanced conditional on Y, E^c, then*

$$R_\gamma(a) \geq (1 - p)/2.$$

Alternatively, suppose Y is coupled to a clean bundle Y_0 for which ℓ is balanced conditional on Y_0, and $Y = Y_0$ outside an event of probability at most p. Then

$$R_\gamma(a) \geq \max\{0, 1/2 - p\}.$$

The assumptions are distinct. Under the first, revealing the label on a leak event independent of the balanced label can attain the bound with forced choice.

Proof. With a balanced posterior, any covered guess has conditional risk $1/2$ and abstention has conditional risk $\gamma \geq 1/2$. In the first case this lower-bounds risk on the clean stratum; risk on leakage is nonnegative. For sharpness use a constant clean bundle and a leak revealing the label, choosing either fixed label on clean observations. For the coupling bound use the same decision randomization on Y_0 and Y. Their losses agree when the bundles agree and differ by at most one otherwise. The risk on Y_0 is at least $1/2$. Nonnegativity gives the stated maximum. Label-dependent contamination can change the balance in the clean stratum, so the first bound cannot be inferred from the coupling alone. □

Proposition 19.3 (Alternative detectability criterion)**.** *Evidence for distinguishability under H_1 requires a preregistered rule $a^\star$, minimum coverage $C_{\min} > 0$, a justified null bound B_0 accounting for the leakage budget, and margin $\eta > 0$. On admissible held-out bundles require $C \geq C_{\min}$ and $R_\gamma(a^\star) \leq B_0 - \eta$ after sampling uncertainty and multiple-comparison corrections. Forced-choice, balanced label permutations, always abstaining and known-leak controls are evaluated under the same loss and coverage protocol. The manuscript reports no campaign satisfying this criterion.*

19.1 Coverage-Sensitive Leakage Bounds

Fix decision randomization independent of the bundle, source and leak event. Let B indicate an executed decision and W a covered error, so $C = \mathbb{E}B$ and $R_\gamma = \mathbb{E}W + \gamma(1 - C)$.

Theorem 19.4 (Sharp Leakage Bounds at Fixed Coverage)**.** *Writing $z_+ = \max\{0, z\}$, the clean-stratum and coupling nulls of Theorem 19.2 respectively give*

$$B_{\mathrm{clean}}(C, p) = \gamma(1 - C) + \tfrac{1}{2}(C - p)_+, \qquad B_{\mathrm{coup}}(C, p) = \gamma(1 - C) + (C/2 - p)_+.$$

Both are sharp for each $C, p \in [0, 1]$, with balanced marginal labels and leak probability at most p, including selective abstention. Minimizing over coverage gives the bounds of Theorem 19.2.

Proof. Clean-stratum balance gives $\mathbb{E}[W\mathbf{1}_{E^c}] = \frac{1}{2}\mathbb{E}[B\mathbf{1}_{E^c}]$. At most p covered mass lies in E, proving the first bound. In the coupling model use the same decision seed on Y and Y_0. On clean input $\mathbb{E}(W_0 - B_0/2) = 0$. The centered variable $W - B/2$ lies in $[-1/2, 1/2]$ and agrees with its clean version whenever the bundles agree. Thus $\mathbb{E}W - C/2 \geq -p$, proving the second bound.

For clean-stratum sharpness, reveal the label on a leak event independent of it, cover $\min\{C, p\}$ there, and cover the remaining mass with a fixed guess in the balanced clean stratum. For coupling sharpness, start with a label-independent clean cover flag of probability C and a fixed covered guess. Replace $\min\{p, C/2\}$ of the wrong covered inputs by revealing inputs, preserving coverage. This replacement may depend on the label. Both bounds are nonincreasing in C for $\gamma \geq 1/2$, so $C = 1$ attains their minima. □

19.2 Exact Inference at the Independent Block

The sampling unit is an independent held-out block and supplies one decision from its entire masked record. Fix the rule on separate calibration data, the noise and exclusion policies, block boundary, and eligible population before inspecting held-out labels. The following model assumes independent identically distributed eligible blocks. Shared participants, runs and other dependence sources belong inside the block or require a sampling model that accounts for that dependence.

For $0 \leq k \leq n$, let $I_\alpha(k, n)$ denote the equal-tail exact binomial interval. The lower endpoint is zero at $k = 0$ and otherwise solves $\Pr_q(\mathrm{Bin}(n, q) \geq k) = \alpha/2$. The upper endpoint is one at $k = n$ and otherwise solves $\Pr_q(\mathrm{Bin}(n, q) \leq k) = \alpha/2$. Set $I_\alpha(0, 0) = [0, 1]$.

Theorem 19.5 (Independent-Block Risk and Coverage Inference)**.** *Let M count covered decisions and K count wrong covered decisions among n independent blocks. Then*

$$M \sim \mathrm{Bin}(n, C), \qquad K \mid M \sim \mathrm{Bin}(M, e), \qquad e = \Pr(W = 1 \mid B = 1) \quad (C > 0).$$

With $I_C = I_{\alpha_C}(M, n)$ and $I_e = I_{\alpha_e}(K, M)$, simultaneous coverage is at least $1 - \alpha_C - \alpha_e$. When $M = 0$ retain $I_e = [0, 1]$ and leave sample conditional error unreported. The minimum and maximum of $\gamma(1 - c) + cx$ over the corners of $I_C \times I_e$ give risk bounds simultaneous with I_C.

A direct joint alternative uses counts $N = (K, M - K, n - M)$ and probabilities $\theta = (Ce, C(1 - e), 1 - C)$. Let f_θ be their multinomial mass and define

$$p_\theta(N) = \sum_{\substack{m_1 + m_2 + m_3 = n \\ f_\theta(m) \leq f_\theta(N)}} f_\theta(m), \qquad \mathcal{Q}_\alpha(N) = \{\theta : p_\theta(N) > \alpha\}.$$

Projecting $\mathcal{Q}_\alpha$ onto $C = \theta_1 + \theta_2$ and $R_\gamma = \theta_1 + \gamma\theta_3$, using infima and suprema, yields a simultaneous exact confidence region with coverage at least $1 - \alpha$.

Suppose a justified leak cap is p_U. For a stochastic audit, let an audit flag D satisfy $\mathbf{1}_E \leq D$ almost surely in the same target population. From J flags in n_D independent audit blocks, the

upper endpoint of $I_{\alpha_D}(J, n_D)$ can supply p_U. With simultaneous coverage endpoints C_L, C_U and risk upper endpoint R_U, certify the preregistered minimum coverage and a fixed positive margin η against the applicable null floor only if

$$C_L \geq C_{\min}, \qquad R_U \leq B_\star(C_U, p_U) - \eta.$$

Under that null this certification has probability at most $\alpha_C + \alpha_e + \alpha_D$ for the rectangle construction, or $\alpha + \alpha_D$ for the joint construction. A deterministic certified cap uses $\alpha_D = 0$.

Proof. The three block outcomes have probabilities $Ce, C(1-e), 1-C$, and factoring their multinomial law proves the binomial laws. Exact binomial tail inversion bounds each failure probability. Conditional interval coverage holds for every M, and a union bound proves the rectangle guarantee. For $C = 0$ the risk is γ independently of e. The risk is affine in each coordinate separately, so both extrema lie at corners.

For the joint method, order the finitely many multinomial masses, including ties. The total mass of outcomes whose inclusive lower mass-rank is at most α is at most α. Thus p_θ is superuniform at the true θ and inversion and projection give the stated guarantee.

The audit upper endpoint bounds $\mathbb{E}D$, which bounds $\Pr(E)$. Its domination of leakage needs a justified audit model. On the simultaneous coverage event, $R_\gamma \leq R_U$, $C \leq C_U$ and $p \leq p_U$. Both null floors are nonincreasing in their arguments, so Theorem 19.4 gives $R_\gamma \geq B_\star(C_U, p_U)$. A positive-margin certification contradicts this inequality. Union bounds complete the error control. Allocate the stated error levels before any family of planned comparisons. □

Proposition 19.6 (Whole-Block Label Permutation Control)**.** *Under conditional exchangeability of source labels across eligible independent blocks, conditional on public records and fixed label counts, fix the guesses and abstentions and permute block labels. If the design fixes exchangeability groups, permute within each group. The fraction of assignments with risk at most the observed risk is an exact lower-tail p-value. Coverage is fixed.*

Proof. The true assignment is uniform on the finite permutation set conditional on the frozen records and design counts. The inclusive lower-tail rank is superuniform by counting assignments. Abstention contributes the same constant for every assignment and each covered error is counted once per block. Conditional exchangeability is the premise of this test. Label-dependent contamination is handled by the applicable leakage bound. □

A four-block synthetic permutation control with two labels of each kind, positive guesses on two fixed blocks and abstentions on two blocks, gives covered-error counts $0, 1, 1, 1, 1, 2$ over the six assignments. At $\gamma = 3/4$ its mean risk is $5/8$, and zero covered errors has lower-tail probability $1/6$. An eight-atom coupling control covers four atoms with two labels of each kind and guesses the positive label there. Correcting one wrong covered guess by label-dependent replacement yields $C = 1/2$, $p = 1/8$, $R_{3/4} = 1/2$. This attains the coupling floor and falls below the clean-stratum floor $9/16$ because the replacement changes the clean conditional balance.

19.3 Gaussian Within-Block Likelihood and GLS

Theorem 19.7 (Stationary AR(1) Likelihood and GLS)**.** *For a block of length $n \geq 1$ let $e_t = y_t - \mu_t$. Assume Gaussian $e_1 \sim N(0, \sigma^2/(1-\rho^2))$ and innovations $e_t - \rho e_{t-1} \sim N(0, \sigma^2)$ independent of each other and of e_1, with $|\rho| < 1$ and $\sigma^2 > 0$. Then*

$$Q_\rho(e) = (1-\rho^2)e_1^2 + \sum_{t=2}^{n} (e_t - \rho e_{t-1})^2,$$

$$\log f(y \mid \mu, \rho, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) + \frac{1}{2}\log(1-\rho^2) - \frac{Q_\rho(y-\mu)}{2\sigma^2}.$$

For two frozen mean models sharing the covariance, the log-density ratio in favor of A is $(Q_\rho(e_B) - Q_\rho(e_A))/(2\sigma^2)$. With full-column-rank design X and fixed common covariance, GLS uniquely minimizes $Q_\rho(y - X\beta)$ at

$$\widehat{\beta} = (X^\top A_\rho X)^{-1} X^\top A_\rho y, \qquad Q_\rho(e) = e^\top A_\rho e.$$

Proof. The residual-to-innovation map is triangular with unit determinant. Multiplication of its independent Gaussian densities gives the formula. For $n \geq 2$ the precision factor A_ρ has endpoints 1, interior diagonal $1 + \rho^2$ and neighboring entries $-\rho$. For $n = 1$ it is $[1 - \rho^2]$. The innovation representation gives positive definiteness and determinant $1 - \rho^2$. Its inverse has entries $\rho^{|i-j|}/(1 - \rho^2)$. Different covariance models retain their separate normalizers and quadratic scales in the log-density difference. With common covariance, differentiating the positive quadratic gives the GLS normal equations and full rank gives uniqueness. □

Freeze the parameters or estimate them on calibration blocks. At $\rho = 1/2$, $\sigma^2 = 1$, residual vectors $(1, 1)$ and $(1, -1/2)$ give log-density ratio $3/8$, while the independent Gaussian approximation gives $-3/8$. Residuals $(2, 1)$ and $(0, 1)$ give stationary log-density ratio -1, while omitting the initial quadratic term gives $1/2$. A constant fit to $(1, 0, 1)$ at this ρ gives GLS estimate $4/5$ and the independent fit $2/3$.

The Gaussian model supplies a score within each block. Independent-block sampling governs its risk inference. The fiber geometry of Paper IX uses a projection and target distances. The projection $(y, z) \mapsto y$ with hidden target $z \in \{0, 1\}$ has fiber diameter one even when two Gaussian laws on y give a discriminating likelihood ratio. With identity projection the fiber diameter is zero, while equal class laws give log-density ratio zero.

20 Reviewer and Adjudicator Independence Architecture

The independence conditions defines independence criteria, and preregistration and packaging binds those criteria into selection and assignment schemas. Without instantiating human reviewers or adjudicators directly, this eliminates ambiguity about the prerequisites required before personnel assignment can proceed.

21 Reviewer Candidate Pool and Assignment Stub Architecture

The provenance reconciliation expands the reviewer and adjudicator layer from workflow criteria into empty typed workflow schemas:

- reviewer candidate pool template;
- adjudicator candidate pool template;
- reviewer assignment stub;
- adjudicator assignment stub;
- reviewer blinding stub.

Each slot remains explicitly null-bearing until execution. The objective is not to simulate human evaluation, but to lock field schemas, null policies, and the preconditions governing eventual human assignment.

22 Execution Pilot Package Architecture

The execution-pilot package is structured as a typed schema family comprising:

- execution session manifest;
- collection session schema;
- reproducibility package skeleton;
- negative-control bundle skeleton;
- matched-intervention bundle skeleton;
- source-target pairing manifest template.

23 Execution Pilot Rehearsal Kit Architecture

The rehearsal protocol specifies a rehearsal verification layer positioned between package readiness and live collection. Its components include:

- execution rehearsal protocol (Appendix C);
- dry-run sequence plan (Appendix C);
- rehearsal event-log schema (Appendix C);
- rehearsal failure modes (Appendix C);
- rehearsal reset policy (Appendix C).

The rehearsal kit validates state transitions, sequence integrity, and record hygiene without collecting empirical data or emitting adjudicative decisions.

24 Reproducibility Skeleton Architecture

The reproducibility shell exists as a slot-bearing structure. It specifies where future preregistration references, dataset references, control-bundle references, analysis-shell references, reviewer bundles, and adjudicator bundles must be anchored.

25 Dataset Population Boundary

The preregistration and packaging and provenance reconciliation enforce a strict boundary among:

- bundle defined;
- bundle frozen;
- bundle populated;
- bundle adjudicatively interpreted.

Only schema definitions are presently realized; downstream population and adjudication remain unexecuted.

26 Dry-Run Event Log Grammar

Rehearsal logging is typed disjointly from empirical evidence logging. The rehearsal grammar permits:

- `rehearsal_initialized`;
- `sequence_step_checked`;
- `sequence_step_blocked`;
- `sequence_step_reset`;
- `rehearsal_not_started`.

None of these event classes may be reinterpreted as human data, reviewer evidence, or adjudicator action.

27 Evidence Admissibility Grammar

The admissibility grammar still recognizes:

- independent reviewer report;
- preregistered human comparator dataset;
- matched intervention dataset;
- negative-control dataset;
- replication log;
- statistical analysis report;
- reviewer conflict disclosure;
- reproducibility package;
- external failure report;
- external inconclusive report.

Invalid substitutes remain central:

- internal runtime artifacts alone;
- self-generated reviewer notes;
- synthetic human data;
- LLM-generated "human" reports;
- undocumented manual judgments;
- unblinded reviewer impressions;
- post-hoc threshold selection;
- cherry-picked examples;
- release inclusion alone;
- Paper 9 publication alone.

28 Failure, Rejection, and Inconclusive Pathways

Section status. protocol specified.

Three major failure classes remain necessary:

1. **protocol invalidation**: preregistration drift, blinding collapse, or conflict violation;
2. **dataset invalidation**: missing controls, invalid substitution, or insufficient provenance;
3. **inconclusive adjudication**: admissible evidence exists but remains undecisive under the declared shell.

29 Statistical Shell Architecture

Section status. protocol specified.

No statistics are executed here. What exists is a declared analysis family layer, each shell binding channel scope, required inputs, required outputs, and non-implications.

30 Adjudication Decision Grammar

Section status. protocol specified.

The future decision family must be typed before it is populated. Paper 10 now recognizes the following decision tokens as grammar only:

- `externally_admissible_but_inconclusive`
- `protocol_invalidated`
- `adjudication_failed`
- `adjudication_passed`
- `reviewer_consensus_absent`
- `adjudicator_decision_withheld`

These tokens specify future admissible states; none are populated here with empirical outcomes.

31 State Grammar for Execution Readiness

The external adjudication protocol distinguishes the following progression states:

32 Controlled Execution Opening

The collection preparation specifies a bounded controlled-execution opening layer positioned between rehearsal and empirical data acquisition: an initial state in which live protocol records may exist as empty shells while collection remains unstarted.

- `execution_opening_policy_defined` means the controlled-execution policy is typed and bound to structured schemas.
- `collection_openable_under_controls` means pre-collection gates pass under explicit contamination and abort constraints.
- `collection_open_pre_evidential` means the system may open a live execution surface while still containing no evidence.

State	Meaning
schema_defined	Field grammar exists, but no operational object family is yet instantiated.
contract_defined	Allowed relation among objects exists, but no package is assembled.
package_ready	The package grammar exists and may be evaluated for completeness.
instantiated_empty	The object family exists locally with null-bearing or empty-bearing fields only.
frozen_local_unfiled	Local freeze semantics exist, but no external filing is claimed.
rehearsal_ready	Rehearsal grammar and rehearsal-kit checks pass, but no live execution shell exists yet.
live_artifacts_instantiated_unpopulated	Live operational shells exist, but every collection-facing structure remains empty.
session_zero_ready	Session-zero initialization is structurally ready, but still not started.
collection_openable_under_controls	Pre-collection admissibility checks pass under contamination and abort controls.
collection_open_pre_evidential	Controlled execution may open as a live state without any populated evidence.
first_collection_openable	Environment integrity and isolated reproducibility audits pass, the collection opening conditions opening gate is structurally open, and no collection-start event has been executed.
collection_started	A real authorized collection-start event exists.
evidence_generated	Real dataset population or admissible log events exist.
adjudication_result_recorded	Reviewer-governed or adjudicator-governed decision records exist.
paper10_final_authorized	Independent post-execution authorization criteria are met.
dry_run_open	Dry-run structure may execute without evidence or adjudication.
externally_filed	A real external filing reference exists.
populated_with_admissible_evidence	The object family is populated by real admissible evidence.
adjudication_result_bearing	Reviewer and adjudicator outputs exist and may support result-bearing sections.

33 Live-But-Unpopulated Record Layer

The collection preparation introduces unpopulated operational shell schemas: session-zero manifests, live collection manifests, empty collection logs, live dataset manifests, reproducibility manifests, empty decision stubs, and evidence-insertion registers.

Each such shell must explicitly preserve:

- `collection_not_started`;
- `evidence_generation_absent`;
- `adjudication_not_open`;
- explicit empty markers such as `empty_dataset = true` where applicable.

Therefore operational shell creation does not constitute dataset population, and empty schemas may never be treated as evidential records.

34 Session-Zero Protocol

Session-zero is the bounded initialization checkpoint that binds local frozen record routing, live collection manifests, empty dataset manifests, empty collection logs, empty decision stubs, contamination controls, abort controls, and the operator checklist. Session-zero may become ready without collection starting.

Session-zero remains distinct from:

- collection start;
- dataset population;
- reviewer action;
- adjudicator action.

35 Pre-Collection Admissibility Gate

Section status. protocol specified.

The pre-collection gate asks whether collection may be opened under controls, not whether collection has already started. This gate checks session-zero readiness, emptiness of live artifacts, emptiness of collection logs, emptiness of decision stubs, absence of evidence, absence of adjudication, and absence of detected contamination.

Passing this gate yields only `collection_openable_under_controls`. It does not yield `collection_started`.

36 Contamination, Abort, and Invalidation Controls

The collection preparation adds an operational invalidation grammar:

- **contamination event**: any unauthorized dataset population, evidence-bearing log entry, or decision-bearing record.
- **abort event**: any operator-controlled termination of the controlled opening before collection start.
- **invalidated session**: any session whose pre-collection boundary, blinding route, or provenance cleanliness is breached.

These control classes bind the transition-ready protocol shells, establishing contamination and abort rules as execution-opening constraints rather than unattached policy tokens.

37 Operator Checklist

The operator checklist governs execution readiness by binding the local frozen record route, live manifest route, empty-log route, empty-dataset state, contamination rules, reset rules, and explicit confirmation that no evidence exists. Until it is complete, collection preparation must not move from `session_zero_ready` to `collection_openable_under_controls`.

38 Collection-Start Boundary

`collection_open_pre_evidential` is not `collection_started`. The boundary is crossed only by a later real operator-authorized collection start event with provenance, contamination-free status, and a first admissible log event.

39 Evidence-Generation Boundary

Evidence generation begins only when a real admissible population event enters the human comparator dataset family, matched intervention family, negative-control family, collection log, reproducibility package, or decision-linked provenance chain. No collection preparation pre-evidential shell instantiated under this specification crosses that boundary.

40 Decision-Record Boundary

The live decision-record stub is an execution-opening shell only. It is not a decision, not an adjudicator action, not a reviewer action, and not an outcome. The protocol preserves that distinction because a populated decision record belongs to a later evidential and adjudicative state family.

41 Transition from Session-Zero to First Admissible Collection

The transition from readiness to collection obeys five ordered steps:

1. session-zero becomes ready;
2. the pre-collection gate opens collection under controls;
3. collection preparation opens a live but still pre-evidential state;
4. a later explicit collection-start command instantiates the first admissible collection event;
5. only then may the first dataset-population event exist.

42 Executable Integrity Verification

The reproducibility protocol adds an executable-integrity verification layer to confirm that the declared implementation environment executes cleanly under reproducible rules. The required audit surfaces include:

- toolchain visibility;
- deterministic dependency installation under the declared package route;
- full build behavior;
- test suite behavior;
- bounded local-start behavior;
- verifier-chain behavior for source-of-truth, comparative closure, collection preparation, rehearsal protocol, and Paper 9 handoff.

This verification step may pass with compatibility warnings such as engine version differences or dependency advisories. A warning-bearing pass remains operationally informative without crossing the boundary into empirical evidence.

43 Clean-Room Reproducibility

The reproducibility protocol also specifies a clean-room reproducibility check asking whether an isolated environment can reinstall dependencies, rebuild the package, rerun key verifiers, and regenerate protocol records without relying on hidden machine-specific state. Clean-room reproducibility may pass with warnings when a documented compatibility route is used, provided the build remains verifiable.

44 Dependency and Build Assumptions

The executable-integrity layer makes dependency and build assumptions explicit:

- the declared execution engine forms part of the reproducibility contract;
- deterministic locked installation remains the preferred setup route;
- a fallback install route may be recorded only as a compatibility surface;
- build success must remain distinct from evidence generation;
- test success must remain distinct from adjudication.

45 Reproducible Record Regeneration

The reproducibility protocol adds a source-driven record-regeneration check verifying whether protocol records can be deterministically regenerated from source definitions and classifying any differences as:

- absent;
- provenance-only;
- semantically stale; or
- blocking.

Because external-adjudication readiness relies on an extensive family of structured schemas, the audit separates semantic drift from ordinary timestamp or lineage variation and rejects manual editing as a substitute for deterministic generation.

46 Local Path Leakage Policy

Clean-room readiness requires that live normative protocol surfaces remain free of machine-specific absolute paths. Historical planning records, when archived, must remain explicitly marked as non-normative.

47 Controlled Execution Opening Audit

The reproducibility protocol audits the controlled-execution opening layer itself without replacing collection preparation, confirming that the live unpopulated records, session-zero gate, and pre-collection gate survive executable verification and clean-room reconstruction.

48 Session-Zero Reproducibility

Session-zero reproducibility requires that an isolated environment reconstruct the session-zero manifest, operator checklist state, and failure-mode assessment without hidden local inputs and without emitting collection-start or evidence-bearing records.

49 Pre-Collection Gate Reproducibility

Pre-collection gate reproducibility requires that an isolated environment reconstruct the pre-collection admissibility result, contamination checklist, and invalidation status while preserving:

- `collection_not_started`;
- `evidence_generation_absent`;
- `adjudication_not_open`.

50 Why Reproducibility Does Not Imply Evidence

Reproducibility shows that the formal and executable surfaces are rebuildable. It does not show that any dataset has been populated, that any human evidence exists, that any reviewer or adjudicator has acted, or that any result-bearing section is licensed.

51 Why Execution Health Does Not Imply Adjudication

Execution health shows that the software and verification pathways operate reliably. It does not imply that collection has started, that evidence has been generated, that reviewer-governed interpretation exists, or that adjudication is open. A healthy execution environment remains pre-evidential until real collection events occur.

52 Transition Conditions for First Admissible Collection

The collection opening conditions may be classified as `first_collection_openable` only when executable integrity and clean-room reproducibility pass verification, normative surfaces remain free of path leakage, record regeneration is non-blocking, collection preparation remains openable, and no uncollected empirical results are claimed.

Even then, the pre-evidential state must still preserve:

- `collection_start_not_executed`;
- `evidence_generation_absent`;
- `adjudication_not_open`;
- `decision_record_absent`.

53 Claim Grammar for Eventual Execution-Era Statements

Any later admissible result-bearing statement must reference:

- preregistration identifier;
- freeze identifier;

- dataset family identifier;
- negative-control family identifier;
- analysis shell identifier;
- reviewer package identifier;
- adjudicator package or decision identifier.

Forbidden grammar remains explicit:

- no leap from comparator similarity to human equivalence;
- no leap from transport-sensitive comparison to identity preservation;
- no leap from admissibility to metaphysical subjecthood;
- no leap from dry-run success to public-release recommendation.

54 Theorem, Proposition, and Protocol Mapping

Family	Dependency set	Formalized now	Requires execution	Additional interpretation premises
Comparator channel contract	BaseCore, Bridge Program, indexed certificate, comparator contracts	yes	no	result claim
Freeze architecture	comparator contracts, independence conditions, preregistration and packaging	yes	no	external filing claim
Reviewer package architecture	independence conditions, preregistration and packaging	yes	no	review outcome
Pilot package architecture	preregistration and packaging	yes	no	evidence claim
Dry-run opening grammar	preregistration and packaging, provenance reconciliation	yes	no	adjudication claim
Human comparator result section	all above + admissible data	no	yes	human equivalence
Adjudication outcome section	all above + reviewer/adjudicator layer + admissible evidence	no	yes	metaphysical closure

55 Evidential Consequences and Interpretive Questions

Paper 10 specifies blind adjudication within the empirical programme. The identification and prediction contracts of Papers IV and VI supply its inputs; interpretation retains independent bridge premises.

Claim family	Possible post-execution movement	Independent bridge requirements
Execution reproducibility	from local/internal to externally reproduced under protocol	consciousness, subjecthood, phenomenality
Human comparator similarity	from unpopulated protocol to measured channel comparison	human equivalence or moral parity
Bridge predicate discrimination	from bridge schema and internal surfaces to adjudicated comparative evidence	phenomenal experience simpliciter
Negative-control survival	from internal controls to externally witnessed controls	theorem confirmation beyond stated assumptions
Failure or inconclusive result	from open uncertainty to documented non-support or bounded inconclusiveness	post hoc reinterpretation

Proposition 55.1 (External evidence is not interpretive closure)**.** *Executed Paper 10 evidence, even if admissible, would not by itself establish human equivalence, metaphysical subjecthood, biological life, phenomenal experience, or moral status.*

Proof. The Paper 10 protocol measures channel comparison, adjudicator behavior, negative controls, provenance, contamination, and statistical distinguishability under frozen rules. None of those objects is identical to the listed interpretive predicates. Moving a claim from internal support to external evidence therefore changes its evidential status, not its semantic type. □

56 Evidence Insertion Map

Human comparator result insertion point. Admissible evidence required: frozen pre-registration record, typed human comparator dataset, negative-control bundle, statistical shell output, reviewer-selection package, reviewer report family.

External adjudication result insertion point. Admissible evidence required: reviewer report set, adjudicator decision record, blinding log, conflict disclosures, reproducibility package, failure and inconclusive pathway disclosures.

Transport-sensitive comparison insertion point. Admissible evidence required: source-target pairing manifest, transition lineage map, axis-specific error budgets, transport negative controls, adjudicator layer.

Instantiated-empty to evidential transition point. Admissible evidence required: local frozen record upgraded by real lock event, real candidate population, real assignment event, real collection session, real decision record population, and real reproducibility package population.

57 Conditions for Empirical Evaluation

Open burdens remain explicit:

- real local frozen preregistration lock absent;
- external filing absent;
- reviewer selection not started;
- adjudicator assignment not started;
- actual blinding assignment absent;
- admissible human comparator dataset absent;
- admissible matched-intervention dataset absent;
- admissible negative-control dataset absent;
- collection sessions not started;
- rehearsal event log not started;
- reproducibility package unpopulated;
- reviewer and adjudicator decision records absent.

58 Execution-Era Unlock Conditions

Before any result-bearing section may become writable, the following must all be true:

1. a frozen preregistration record exists;
2. any required external filing state is real and documented;
3. candidate pools are populated by real identities with provenance;
4. reviewer selection is instantiated;
5. adjudicator assignment is instantiated;
6. blinding assignment is instantiated;
7. admissible datasets and control bundles are populated;
8. collection logs and execution sessions are populated by real events;
9. reproducibility package slots are populated;
10. decision grammar is instantiated by real evidence.

59 Stronger Execution-to-Result Transition Map

The transition from collection preparation controlled opening to a result-bearing protocol has six distinct unlock layers:

1. **Instantiation layer**: empty records, pools, stubs, and rehearsal kit exist.
2. **Controlled opening layer**: live operational shells, session-zero, and pre-collection controls exist while collection still remains not started.
3. **Population layer**: real humans, real sessions, and real evidence populate those empty structures.
4. **Admissibility layer**: controls, provenance, blinding, and declared shells survive audit.
5. **Decision layer**: reviewer and adjudicator records become populated by real independent action.
6. **Empirical reporting layer**: only then may result-bearing Paper 10 sections be populated.

60 Conditions for Comparative Conclusions

Paper 10 specifies the normative protocol prior to data collection and therefore withholds empirical conclusions until external evidence-bearing surfaces exist. The text specifies the machinery of freezing, selection, packaging, dry-run, provenance closure, instantiated-empty records, and rehearsal-facing protocol oversight, while prohibiting any conclusion whose truth depends on uncollected evidence.

61 Execution Pathway to External Evidence

The operational execution sequence proceeds through nine ordered steps:

1. populate the local frozen record under preregistration lock;
2. populate reviewer and adjudicator pools with verified disclosures;
3. instantiate actual reviewer, adjudicator, and blinding assignments;
4. validate dry-run, rehearsal, and controlled-opening integrity;
5. issue a real collection-start event under collection preparation controls;
6. populate admissible datasets, collection logs, and control bundles;
7. run declared statistical shells;
8. bind reviewer and adjudicator outputs;
9. only then populate result-bearing sections.

62 Boundary to Metaphysical Consequence Analysis

Metaphysical consequence analysis remains outside the authority surface of this document. Even later successful execution-era evidence would not automatically license metaphysical subjecthood, moral status, or cross-substrate identity preservation.

63 Empirical Evaluation Conditions

63.1 Human Comparator Results

Admissible evidence required. Frozen preregistration record required. Execution records required. Negative controls required. Statistical shell required. Reviewer package required.

63.2 External Adjudication Results

Admissible evidence required. Independent reviewer reports required. Conflict disclosures required. Blinding log required. Adjudicator decision record required.

63.3 Comparative Consequence Claims

Admissible evidence required. Execution-era records required. Adjudication-facing evidence required. These records support bounded empirical comparisons; identity preservation and normative interpretations require independent bridge premises.

64 Open Interpretive Questions

The association of operational certificates with phenomenal experience remains open. Comparative human interpretation, physical identity and normative status require independent bridge premises and evidence.

65 External Interface Admission Before Biological Adjudication

The experimental interface contract in Paper IV is an admission prerequisite, not a new support label. An external submission supplies its population/preparation, independent blocks, sensory/motor and measurement maps, units and noise calibration, matched sham, causal disconnection preserving observation, restoration control, frozen blinded decoder, rival information/cost and causal identification assumptions. Missing maps are recorded as insufficient evidence; invented biological observations are not admissible.

The forced-choice/abstention loss of this paper remains unchanged. It concerns labels and cannot replace the model-specific falsation contrast. For the latter, the Paper IV blockwise rule bounds false rejection by $\min(1, \alpha + \beta + p)$ only when the claimed audit failure probability is backed by a coupling to the specified blind reference design. An observed audit pass rate alone is insufficient. Calibration failure, leakage, dependence between purported units and off-target disconnection effects have separate status fields. A valid rejection bears on the realized model and its observation/causal contract; it does not refute an abstract theorem whose hypotheses have not been instantiated. Acceptance here supplies no empirical support until an actual independent submission satisfies the existing evidence and protocol oversight gates. External support remains uncertified.

Admission of the new constructive certificates. Paper VIII's native infinite-dimensional member proves nonemptiness of its instrumented indexed class relative to a fully specified causal/computational grammar. Paper I supplies metric-sector product gaps and a conditional protected-actuator threshold. Neither result changes this adjudicator's loss, coverage, blinding, leakage or external-evidence requirements. In a submission using the indexed forecast, admit same-information/cost emulators as an explicit sensitivity control; if they match the candidate, the raw comparative margin is nonpositive. In a wedge submission, declare factor diameters, distortion constants and actual energy calibration rather than borrowing a mathematical actuator's

units. Independent data, method review and replication remain unexecuted unless a submission documents them; `external_support_certified=false` remains the internal boundary.

66 Interpretation and Open Research Links

Historical alias. External Adjudication of Bridge-Formalized Machine Subjectivity. The alias is retained for bibliographic continuity; the technical title states the demonstrated object.

Formal anchor. Results 19.2 and 19.4.

Proposed interpretation. Blind comparative success could support a bounded operational interpretation. It is not a definition or direct observation of experience.

Additional obligations. The analytic risk bounds require their balance, leakage, independence and coverage assumptions; collection and external filing remain unexecuted unless sourced.

Differential tests and evidence. Run forced-choice, abstention, permutation and known-leak controls with justified sampling uncertainty and independent review.

A Preregistration and Blind Selection Contracts

Table 1: Preregistration and Blind Selection Contracts.

Contract	Specification and evidence conditions
Frozen Preregistration Record Template	Schema family: frozen preregistration record specification, governed by the preregistration freeze, amendment, and lock-manifest policies. Admissible evidence required before population: actual local lock and, where required by protocol, real external filing reference.
Human Comparator Preregistration Template	Schema family: human comparator channel preregistration specification. Admissible evidence required before population: channel-level freeze and later dataset lineage.
Reviewer Candidate Schema	Schema family: reviewer candidate profile and conflict-governed selection workflow. Admissible evidence required before population: real candidate package and conflict disclosures.
Adjudicator Candidate Schema	Schema family: adjudicator candidate profile and authority-bounded assignment workflow. Admissible evidence required before population: real assignment package and decision-authority boundary.
Blinding Assignment Template	Schema family: blinding assignment manifest specification. Admissible evidence required before population: authorized assignment route and later unblinding event if one exists.
Local Frozen Record Example	Schema family: local frozen preregistration record specification. Admissible evidence required before result use: real local lock event and, where required by route, external filing provenance.
Reviewer Candidate Pool Template	Schema family: reviewer candidate pool specification. Admissible evidence required before result use: real reviewer identities, conflict disclosures, and selection events.
Adjudicator Candidate Pool Template	Schema family: adjudicator candidate pool specification. Admissible evidence required before result use: real adjudicator identity, assignment provenance, and decision-authority declaration.
Blinding Assignment Stub	Schema family: pre-evidential blinding assignment and reviewer-blinding structural stubs. Admissible evidence required before result use: real assignment route and real unblinding trace if one later exists.

B Session, Collection, and Provenance Records

Table 2: Session, Collection, and Provenance Records.

Contract	Specification and evidence conditions
Execution Pilot Package Manifest	Schema family: execution pilot package manifest specification. Admissible evidence required before population: real session manifests and later populated bundle references.
Dataset Contract Summary	Schema family: matched-intervention and negative-control dataset contracts, together with the pilot matched-intervention bundle specification. Admissible evidence required before result use: populated admissible datasets.
Statistical Shell Summary	Schema family: preregistered statistical analysis shell specification. Admissible evidence required before result use: shell execution output on real admissible inputs.
Adjudication Decision Grammar Summary	Decision tokens retained as grammar only: `externally_admissible_but_inconclusive`, `protocol_invalidated`, `adjudication_failed`, `adjudication_passed`, `reviewer_consensus_absent`, `adjudicator_decision_withheld`.
Execution Session Template	Schema family: execution session manifest specification. Admissible evidence required before result use: real collection session and real session lineage.
Empty Collection Log Schema	Schema family: pre-evidential unpopulated collection-log schema. Admissible evidence required before result use: populated log entries from real collection events.
Empty Decision Record Template	Schema family: pre-evidential decision-record structural template. Admissible evidence required before result use: real reviewer and adjudicator action, not grammar only.
Reproducibility Manifest Example	Schema family: unpopulated reproducibility package manifest specification. Admissible evidence required before result use: populated package slots, dataset references, shell outputs, and reviewer-facing bundles.
Evidence Insertion Manifest	Schema family: evidence insertion manifest specification. Admissible evidence required before result use: actual record references, actual decision references, and populated provenance chains.
Session-Zero Protocol and Checklist	Schema family: session-zero protocol and operator-checklist specification. Admissible evidence required before result use: a later real collection-start event and later populated log entries.
Live-But-Unpopulated Record Map	Schema family: live pre-evidential collection manifest, live human-comparator dataset manifest, and parallel collection-protocol live operational shells. Record scope: live operational shell only. Additional evidence is required for: populated dataset, evidence, or adjudication.

C Integrity, Reproduction, and Opening Conditions

Table 3: Integrity, Reproduction, and Opening Conditions.

Contract	Specification and evidence conditions
Negative Control Bundle Summary	Schema family: pilot negative-control bundle specification. Admissible evidence required before result use: populated bundle and control admissibility record.
Rehearsal-Kit Map	Schema family: execution-pilot rehearsal kit and dry-run sequence specification, governed by rehearsal failure-mode and reset policies. Admissible evidence required before result use: none at rehearsal time, and no rehearsal record may be promoted into evidence.
Dry-Run Event Log Grammar	Schema family: rehearsal event-log schema. Allowed event classes remain rehearsal-only and do not imply human data, reviewer action, or adjudicator action.
Controlled Execution Opening Policy	Schema family: controlled execution-opening policy specification. Admissible evidence required before result use: none at opening time, and no controlled-opening record may be promoted into evidence by itself.
Pre-Collection Gate and Contamination Controls	Schema family: pre-collection admissibility gate, acceptance-criteria, and contamination-checklist specification. Admissible evidence required before result use: a later contamination-free collection-start event and later admissible dataset population.
First Admissible Collection Unlock Map	Schema family: first admissible collection unlock protocol and gate rules. Admissible evidence required before result use: actual reviewer selection, actual adjudicator assignment, actual blinding assignment, operator authorization, collection-start command, and first populated collection-log event.
Executable Integrity Verification Audit	Schema family: executable-environment health and verifier-chain audit record. Record scope: executable health and verifier-chain status only. Additional evidence is required for: evidence, collection start, or adjudication.
Clean-Room Reproducibility Audit	Schema family: clean-room rebuildability and reproducibility audit record. Record scope: bounded rebuildability only. Additional evidence is required for: external adjudicative support, empirical support, or completed empirical confirmation.
Local Path Leakage Audit	Schema family: path-leakage and self-containment classification audit. Record scope: leak classification only. Historical snapshots remain preserved; path sanitization concerns the present distribution.
Record Regeneration Audit	Schema family: deterministic record regeneration and drift-classification audit. Record scope: regeneration and drift classification only. Forbidden interpretation: result-bearing evidence insertion.
First Collection Opening Criteria	Schema family: first admissible collection opening criteria, collection-start policy, opening gate, and opening manifest. Record scope: first-collection opening criteria only. Forbidden interpretation: executed collection start, evidence generation, or adjudication.

Authorial Note and Methodological Provenance

The conception, theoretical direction, mathematical architecture, development, and integration of the QICN research programme were undertaken by a single independent researcher, Johnny Andrey Pérez Ramírez, over more than one year. This account of provenance is supplied by the author. Several artificial intelligence systems and formal verification tools assisted with technical drafting, proof exploration, consistency checks, and mechanized kernels under his direction. The author retains intellectual responsibility for the programme, its claims, and any remaining errors. Analytical manuscript proofs, machine-checked kernels, and internal tests have distinct scopes; independent mathematical review and empirical validation remain separate obligations. `external_support_certified=false`.